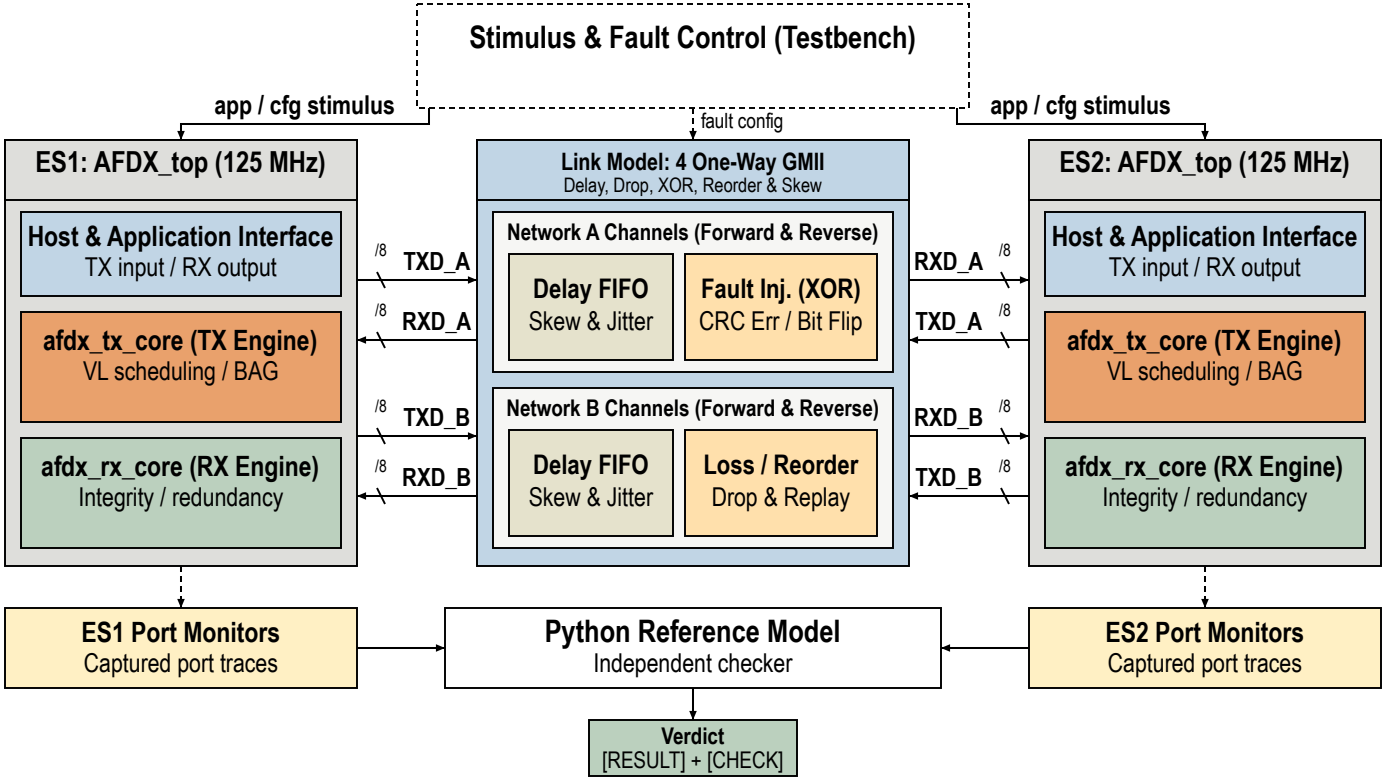

Stimulus & Fault Control (Testbench)
app / cfg stimulus
fault config
app / cfg stimulus
ES1: AFDX_top (125 MHz)
Host & Application Interface
TX input / RX output
afdx_tx_core (TX Engine)
VL scheduling / BAG
afdx_rx_core (RX Engine)
Integrity / redundancy
Link Model: 4 One-Way GMII
Delay, Drop, XOR, Reorder & Skew
Network A Channels (Forward & Reverse)
Delay FIFO
Skew & Jitter
Fault Inj. (XOR)
CRC Err / Bit Flip
Network B Channels (Forward & Reverse)
Delay FIFO
Skew & Jitter
Loss / Reorder
Drop & Replay
/8
TXD_A
/8
RXD_A
/8
TXD_B
/8
RXD_B
/8
RXD_A
/8
TXD_A
/8
RXD_B
/8
TXD_B
ES2: AFDX_top (125 MHz)
Host & Application Interface
TX input / RX output
afdx_tx_core (TX Engine)
VL scheduling / BAG
afdx_rx_core (RX Engine)
Integrity / redundancy
ES1 Port Monitors
Captured port traces
Python Reference Model
Independent checker
ES2 Port Monitors
Captured port traces
Verdict
[RESULT] + [CHECK]